Car Value Estimator

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Section “D”

2. Abstract of the Project

- **Problem Statement:**
  The objective of this project is to develop a machine learning model to estimate car prices based on various attributes. Accurate price estimation can benefit buyers and sellers by providing fair market value insights and can aid in financial planning for vehicle purchases.
- **Prediction Goal:**
  We aim to predict the price of a car based on attributes such as make, model, year, mileage, and other relevant features

3. Source of the Data

Source of the Data:
We will use the Kaggle Car Prices Dataset as it provides comprehensive data on car prices and various features.

First Few Records of the Dataset:

Car Make	Model	Year	Mileage	Price
Toyota	Camry	2020	25000	25000
Honda	Accord	2018	30000	22000
Ford	Focus	2019	15000	18000
BMW	X5	2021	10000	45000

4. Description of Each Attribute

- **Car Make**: The manufacturer of the car (e.g., Toyota, Honda, Ford).
- **Model**: The specific model of the car (e.g., Camry, Accord, Focus).
- **Year**: The year the car was manufactured.
- **Mileage**: The total distance the car has traveled, usually measured in miles or kilometers.
- **Price**: The selling price of the car

5. Learning Methodology and Algorithm

- **Learning Methodology:**
  We will use supervised learning methodology as we have labeled data (car prices) and will be training our model to predict these labels.
- **Algorithm:**
  We will use regression algorithms for this project, specifically:
- **Linear Regression:** To start with a basic approach and establish a baseline.
- **Random Forest Regression:** To handle more complex relationships between features and prices.
- **Gradient Boosting Machines (GBM):** To improve performance through boosting techniques.

Thank You